

Advanced (Carolina Reaper)

- Q1.** Graph the system of inequalities: $x + y \geq 2$ and $x - y \leq 1$. What is the shape of the solution region?
- (A) Triangle
(B) Quadrilateral
(C) Pentagon
(D) Circle
(E) Parallelogram
- Q2.** A book costs 15 and a pen costs 5. Graph the system of inequalities representing the total cost of buying x books and y pens, given that the total budget is 50.
- (A) $x + y \leq 10$
(B) $15x + 5y \leq 50$
(C) $x - y \geq 5$
(D) $3x + 2y \geq 20$
(E) $2x - y \leq 10$
- Q3.** A company produces two types of products: x and y . The profit from x is 20 per unit and from y is 15 per unit. The graph of the system of inequalities representing the total profit is
- (A) $20x + 15y \leq 100$
(B) $15x + 20y \leq 50$
(C) $x + y \leq 10$
(D) $2x - y \geq 5$
(E) $3x + 2y \geq 20$
- Q4.** Tom has 25 to spend on video games and snacks. Video games cost 10 each and snacks cost 5 each. Graph the system of inequalities representing the total cost of buying x video games and y snacks.
- (A) $10x + 5y \leq 25$
(B) $5x + 10y \geq 20$
(C) $2x - y \leq 10$
(D) $x + y \geq 5$
(E) $3x + 2y \leq 15$
- Q5.** A bakery sells two types of cakes: x and y . The system of inequalities representing the total number of cakes sold is $x + y \leq 20$ and $x - y \geq 5$. What is the minimum number of cakes sold?
- (A) 5
(B) 10
(C) 15
(D) 20
(E) 25
- Q6.** A student has 12 hours to study for two exams. The system of inequalities representing the time spent studying for each exam is $x + y \leq 12$ and $x - y \geq 2$. What is the maximum time spent studying for the first exam?
- (A) 5 hours
(B) 6 hours
(C) 7 hours
(D) 8 hours
(E) 9 hours
- Q7.** A store sells two types of shirts: x and y . The profit from x is 10 per unit and from y is 8 per unit. The system of inequalities representing the total profit is $10x + 8y \geq 50$ and $x - y \leq 5$. What is the minimum profit?
- (A) 50
(B) 60
(C) 70
(D) 80
(E) 90
- Q8.** A company produces two types of products: x and y . The cost of producing x is 5 per unit and the cost of producing y is 3 per unit. The system of inequalities representing the total cost is $5x + 3y \leq 50$ and $x + y \geq 10$. What is the minimum cost?
- (A) 30
(B) 35
(C) 40
(D) 45
(E) 50

Open Ended Questions (FRQ)

- Q9.** Graph the system of inequalities: $2x + y \geq 4$ and $x - 2y \leq 3$. Identify the overlapping solution region and interpret the feasible region.

Q10. A student has 15 hours to study for three exams. The system of inequalities representing the time spent studying for each exam is $x + y + z \leq 15$ and $x - y \geq 2$ and $y - z \leq 3$. Graph the system and identify the minimum time spent studying for the first exam.

Chef's Tips

- Tip 1: Ensure students understand the concept of graphing multiple inequalities on the same coordinate plane.
- Tip 2: Emphasize the importance of identifying overlapping solution regions and interpreting feasible regions.
- Tip 3: Encourage students to use real-life scenarios to understand the application of systems of linear inequalities.